数学花园
探秘
MATHEMATICS GARDEN

2022 Spring Cup
Mathematical Olympiad

PRELIMINARY ROUND

Date: 28 January 2023

Time Given: 1 hour 30 minutes

Level: Secondary Advanced

Name: _____

Instructions to Candidates

1. Do not open the booklet until you are told to do so.
2. Answer ALL 20 questions.
3. Write your answers in the answer sheet provided.
4. No steps are needed to justify your answers.
5. Questions 1-4 are worth 8 marks each.
6. Questions 5-8 are worth 10 marks each.
7. Questions 8-11 are worth 12 marks each.
8. Questions 12-13 are worth 15 marks each.
9. No marks will be deducted for wrong answers.
10. No marks will be given for unanswered questions.

Part 1 Short Answer Questions I (8 Marks Each, Total 32 Marks)

1. Find the value of the expression $\sqrt{3\left(\sqrt{2}+\sqrt{3}+\sqrt{5}\right)\left(3\sqrt{2}+2\sqrt{3}-\sqrt{30}\right)}$.

Answer: _____

2. If $a^2 + b^2 = 100$, and $(a - b)^2 \geq 200$, calculate the value of $a^4 + b^4$.

Answer: _____

3. If $a^2 + 4a + 1 = 0$, and $\dfrac{a^4+ma^2+1}{2a^3+ma^2+2a} = -10$, find the value of m.

Answer: _____

4. If $a=\sqrt{\dfrac{\sqrt{3}}{6}+\dfrac{1}{48}}-\dfrac{\sqrt{3}}{12}$, find the value of $\left(a^2+\sqrt{a^4+a+2}\right)^2$.

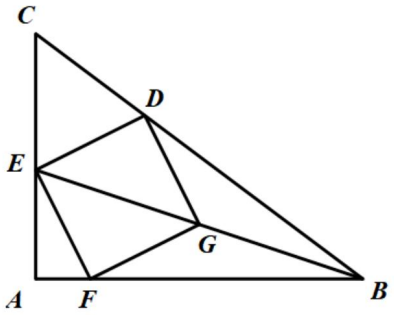
Answer: _____

Part 2 Short Answer Questions II (10 Marks Each, Total 40 Marks)

5. Find the value of integer m that make the result of expression $n^3-14n^2+56n-64$ is prime number.

Answer: _____

6. As shown in the figure, three vertices of square *DEFG* lie on the three sides of right triangle $\triangle ABC$, and points *E*, *G*, and *B* are collinear. Given that $\angle A = 90°$, $AC = 9$, and $AB = 12$, find the area of square *DEFG*.



Answer: _____

7. Let *k* be an integer. The equation in *x*, $\dfrac{(k+4)x^2+(k+2)x-8}{(k+1)x-2} = kx+1$ has real solutions. If the absolute value of the product of all distinct real roots corresponding to all different integers *k* equals $\dfrac{m}{n}$ (where *m* and *n* are coprime positive integers), find the value of $m + n$.

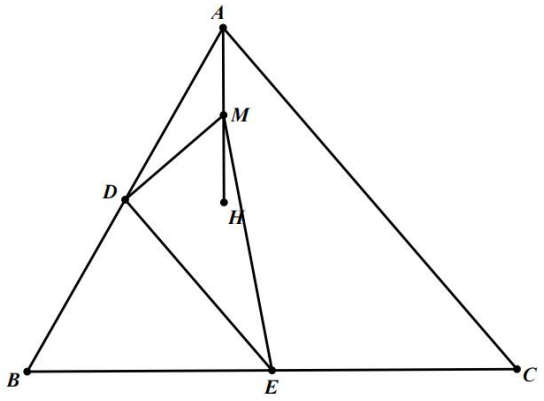
Answer: _____

8. If a positive integer *a* is given, and there are exactly 2022 pairs of positive integers (*b*, *c*)such that $2x^2+3xy+y^2+ax+by+c$ can be factored into the product of two linear factors, find the sum of all possible values of *a*.

Answer: _____

Part 3 Short Answer Questions III (12 Marks Each, Total 36 Marks)

9. As shown, H is the orthocenter of triangle ABC. Points D, E, M are the midpoints of AB, AC and AH respectively. If $\angle BAC = 70°$ and $\angle ACB = 50°$, find the value of $\angle DME$.

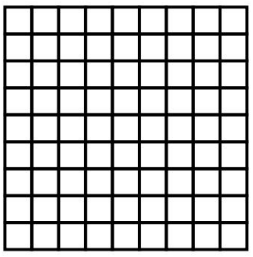


Answer: _____

10. Given the function $y = \left(\sqrt{a^2 - x^2} + m\right)\left(\sqrt{a + x} + \sqrt{a - x} + n\right)$ where m, n are natural numbers and a is a perfect square, the difference between the maximum and minimum values of the function is 2022. Find the value of $m + n + a$.

Answer: _____

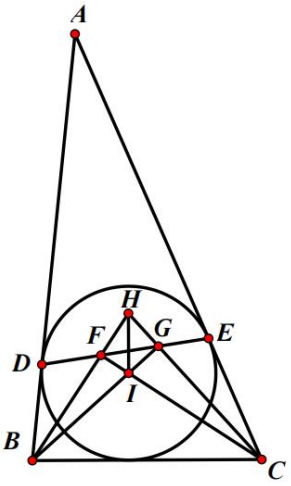
11. In the 9×9 grid shown on the right, we want to select nine 1×1 cells to color black, leaving all other cells white, such that each row, each column, and each diagonal contains exactly one black cell, and the entire figure remains axis-symmetric. Find the number of possible selections.



Answer: _____

Part 4 Full Length Questions III (15 Marks Each, Total 30 Marks)

12. As shown in the figure, the incircle of $\triangle ABC$ touches sides AB and AC at points D and E respectively. Let I be the incenter. The extensions of BI and CI intersect segment DE at points F and G respectively. The extensions of BF and CG meet at point H. Given that $\frac{BC}{HI} = \sqrt{15}$, find the value of $\frac{AD}{DE}$.



13. As shown in the figure, three vertices of the right triangle $\triangle AOB$ lie on the graph of the parabola $y = x^2 + 2x$, with the right-angled vertex at the origin. Let P be the foot of the perpendicular from the right angle to the hypotenuse AB. Find the sum of the maximum and minimum distances from the vertex of the parabola to point P.